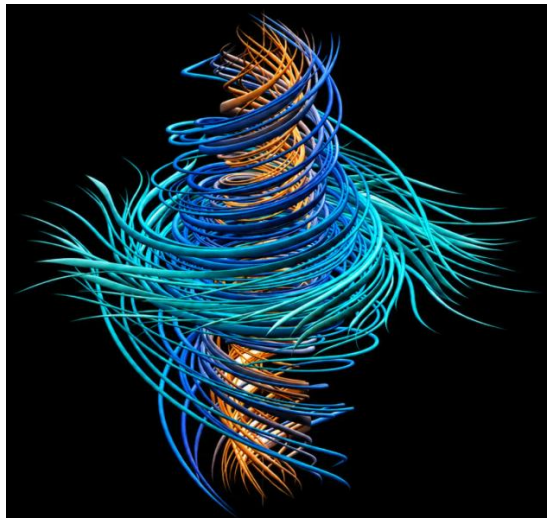
Physics Foundation Model

物理基座模型

求解PDE——我们的野心越来越大



$$\mathcal{F}(u;a)=0$$ 求解

传统求解　　数值计算（FEM…）　　非常复杂

机器学习　　PINNs …　　解1个PDE

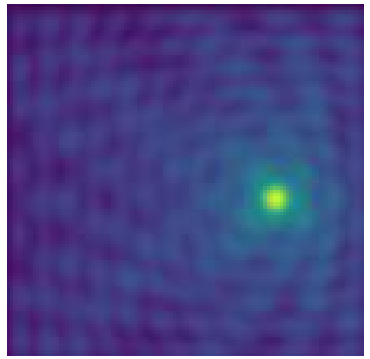
算子学习　　$a(x) \xrightarrow{\mathcal{G}_\theta} u(x,t)$　FNO DeepONet …　　解1族PDE

物理基座模型??　　覆盖多种PDE

$u(x,t)$

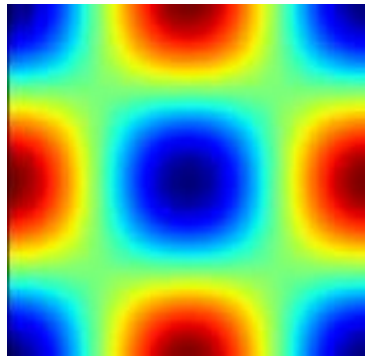
物理基座模型

一种甚至一个PDE就要单独训练模型

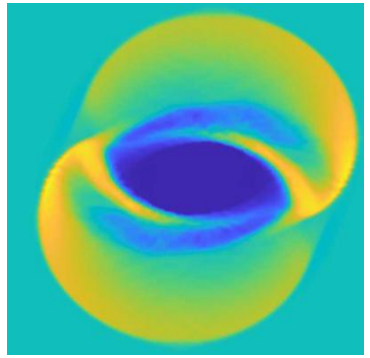


$Model_{\mathrm{Wave}}(x,t)$

$Model_{\mathrm{Darcy}}(x,t)$

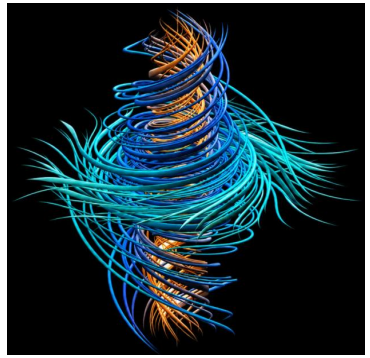


Physics Foundation Model解决所有问题！



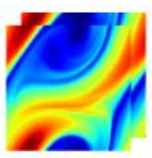
$Model_{\mathrm{MHD}}(x,t)$

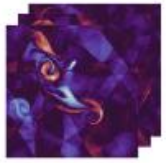
$Model_{\mathrm{NS}}(x,t)$

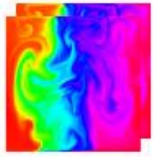


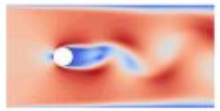
Foundation Model

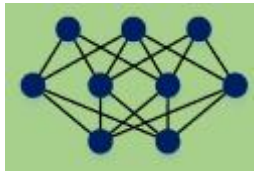
Multiple PDE datsets







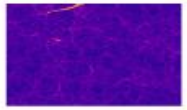


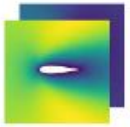


微调下游任务



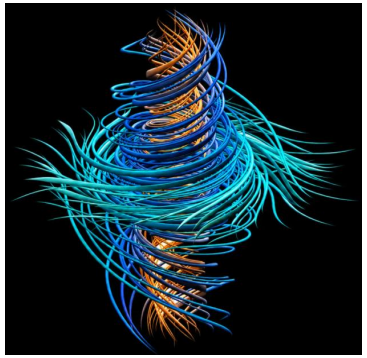


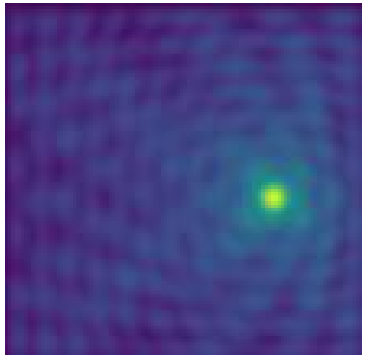


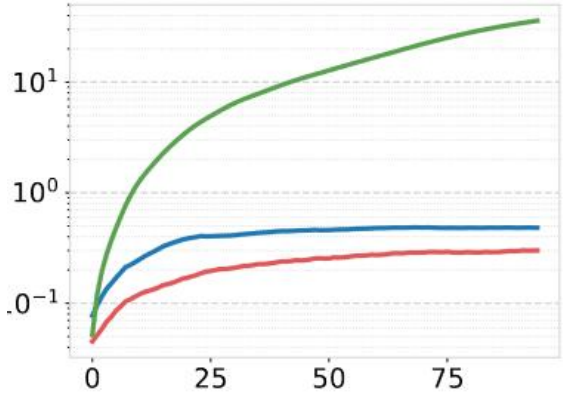


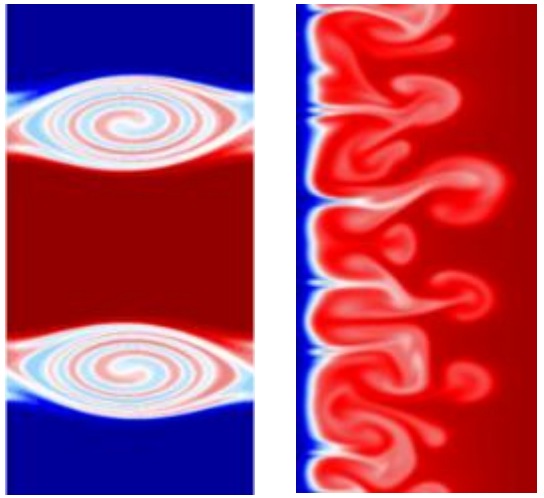
物理基座模型：实现痛点

Data	Representation	Dynamics	Generalization
数据昂贵、异构	2D、3D、tensor、vector	长时累积误差、多尺度	泛化到未见物理？



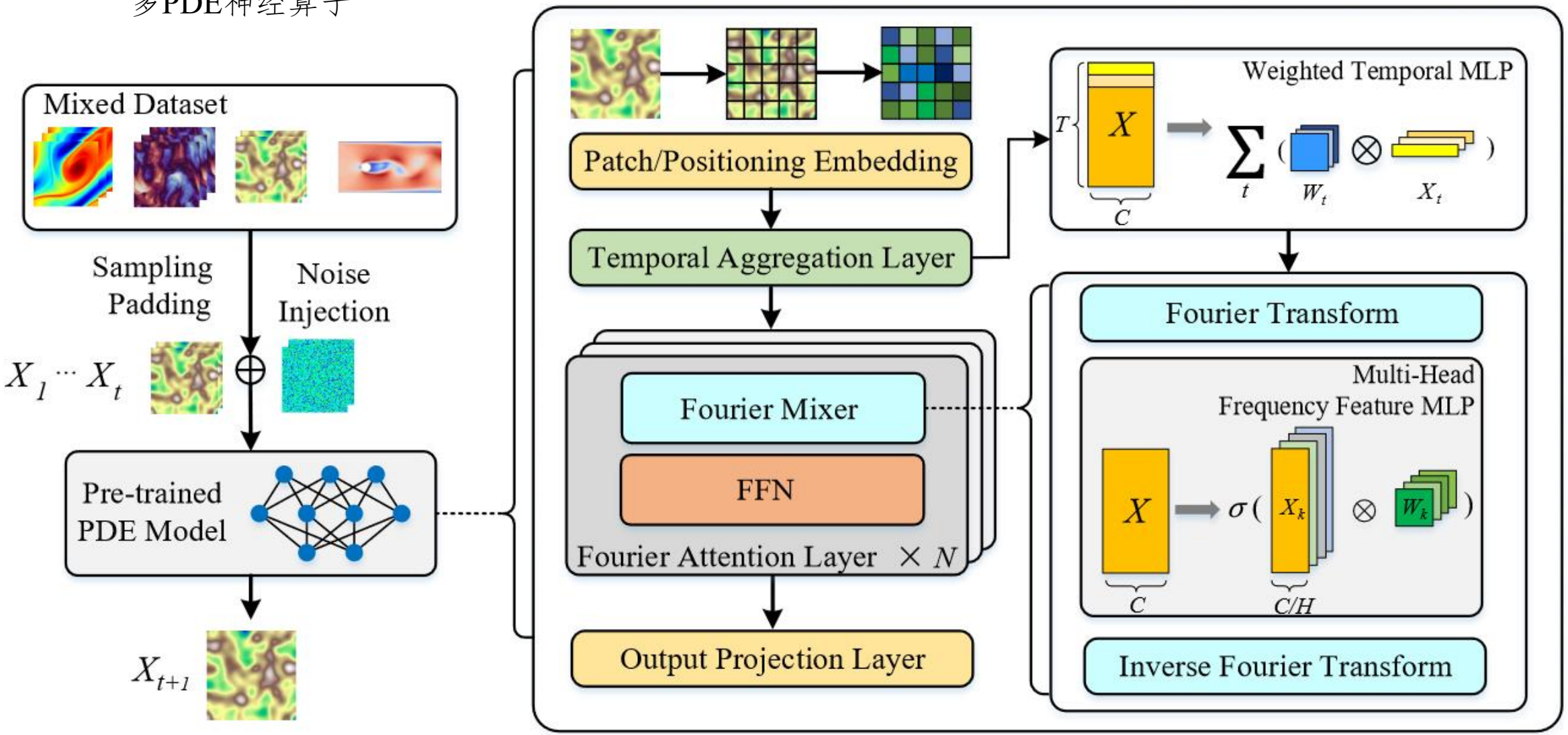






DPOT:Auto-Regressive Denoising Operator Transformer

多PDE神经算子



DPOT:Auto-Regressive Denoising Operator Transformer

Patch/position embedding

$$Z_p^t = \mathcal{P}(\boldsymbol{u}^t + \boldsymbol{p}^t)$$

多帧的信息聚合，隐式识别 PDE 类型和参数

$$\boldsymbol{z}_{\text{agg}} = \sum_t W_t \cdot \boldsymbol{z}_p^t e^{-i\boldsymbol{\gamma} t}$$

使用傅里叶提取空间特征

$$z^l \xrightarrow{\mathcal{F}} \hat{z}^l(k)$$

$$\hat{z}^{l+1} = W_2\sigma(W_1\hat{z}^l + b_1) + b_2$$

$$z^{l+1} = \mathcal{F}^{-1}[W_2\sigma(W_1\mathcal{F}(z^l) + b_1) + b_2]$$

分辨率统一128*128（2D数据）：上/下采样

通道数统一：常数填充

几何适配：mask通道

DPOT:Auto-Regressive Denoising Operator Transformer

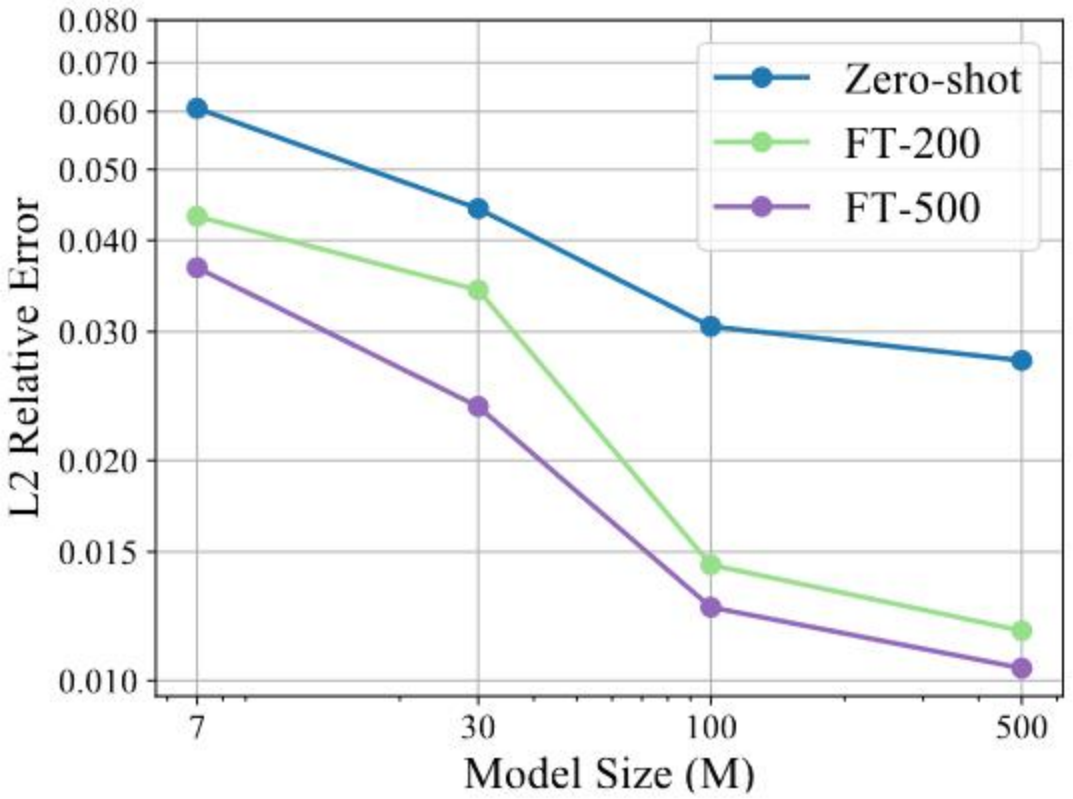
DPOT: Auto-Regressive Denoising Operator Transformer for Large-Scale PDE Pre-Training

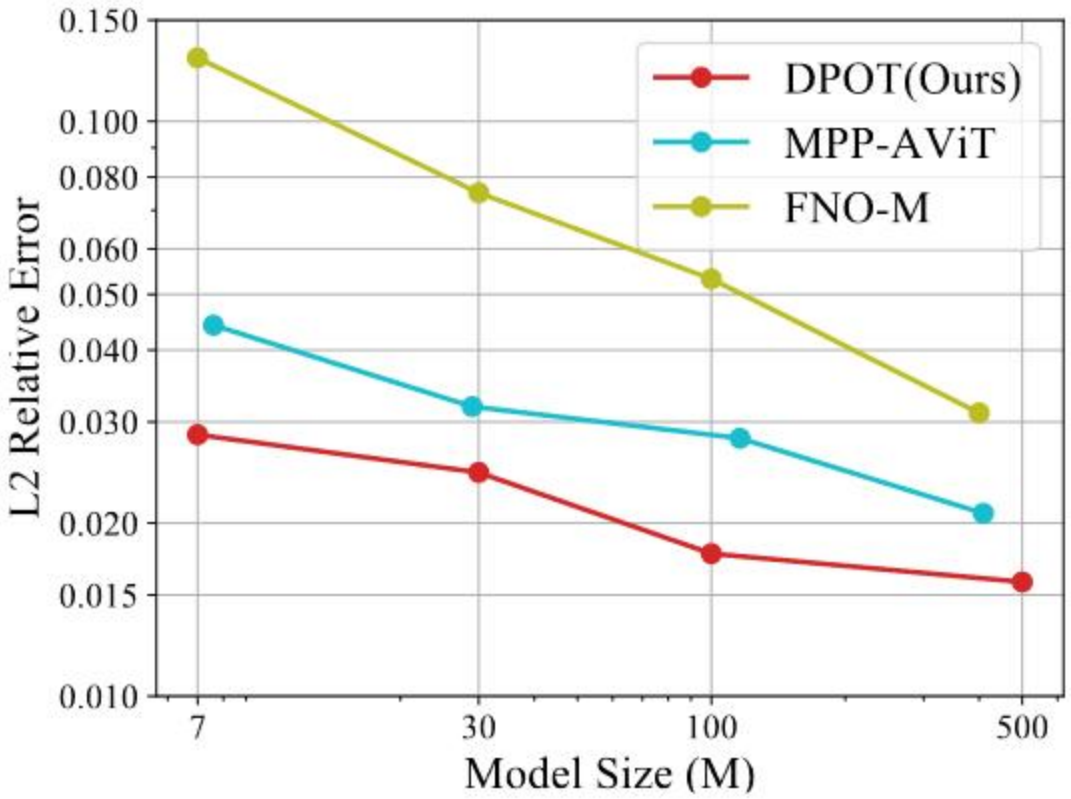
L2RE	Params	FNO-ν			PDEBench CNS-(η, ζ),DR,SWE								PDEArena		CFDBench
Subset	–	1e-5	1e-4	1e-3	1,0.1	1,0.01	M1	0.1,0.1	0.1,0.01	M0.1	DR	SWE	NS	NS-cond	–
Small Model															
FNO	0.5M	0.156	0.0834	0.0128	0.098	0.096	0.097	0.360	0.170	0.265	0.12	0.0044	0.0912	0.319	0.00761
UNet	25M	0.198	0.119	0.0245	0.334	0.291	0.313	0.569	0.357	0.463	0.0971	0.0521	0.102	0.337	0.209
FFNO	1.3M	0.121	0.0503	0.0099	0.0212	0.052	0.0366	0.162	0.0452	0.104	0.0571	0.0116	**0.0839**	0.602	0.00714
GK-T	1.6M	0.134	0.0792	0.0098	0.0341	0.0377	0.0359	0.0274	0.0366	0.0320	0.0359	0.00692	0.0952	0.423	0.0105
GNOT	1.8M	0.157	0.0443	0.0125	0.0325	0.0420	0.0373	0.0228	0.0341	0.0285	0.0311	0.00678	0.172	0.325	0.00877
Oformer	1.9M	0.1705	0.0645	0.0104	0.0417	0.0625	0.0521	0.0254	0.0205	0.0229	0.0192	0.00717	0.135	0.332	0.0102
FNO-m	7M	0.116	0.0922	0.0156	0.151	0.108	0.130	0.230	0.076	0.153	0.0321	0.00912	0.210	0.384	0.0274
MPP-Ti	7M	–	–	–	–	–	0.0442	–	–	0.0312	0.0168	0.0066	–	–	–
MPP-S	30M	–	–	–	–	–	0.0319	–	–	0.0213	**0.0112**	**0.0024**	–	–	–
Ours-Ti	7M	0.0976	0.0606	**0.00954**	0.0173	0.0397	0.0285	0.0132	0.0220	0.0176	0.0321	0.00560	0.125	0.384	0.00952
Ours-S	30M	**0.0553**	**0.0442**	0.0131	**0.0153**	**0.0337**	**0.0245**	**0.0119**	**0.0187**	**0.0153**	0.0379	0.00657	0.0991	**0.316**	**0.00696**
Pre-trained															
MPP-L	400M	–	–	–	–	–	0.0208	–	–	0.0147	**0.0098**	0.00220	–	–	–
Ours-S	30M	0.0553	0.0442	0.0131	0.0153	0.0337	0.0245	0.0118	0.0188	0.0153	0.0379	0.00657	0.0999	0.316	0.00696
Ours-M	122M	0.0409	0.0285	0.00474	0.0116	0.0238	0.0177	**0.00866**	0.0129	0.0108	0.0292	0.00290	0.0812	0.276	0.00752
Ours-L	500M	0.0550	0.0274	0.00528	0.0100	0.0216	0.0158	0.00872	0.0115	0.0101	0.0232	0.00233	0.0798	0.240	**0.00650**
Ours-H	1.03B	**0.0174**	**0.0131**	**0.00229**	**0.00961**	**0.0180**	**0.0138**	**0.00847**	**0.0105**	**0.00948**	0.0191	**0.00199**	**0.0379**	**0.213**	0.00749
DPOT-FT															
T-200	7M	0.0511	0.0431	0.00729	0.0136	0.0238	0.0187	0.0168	0.0145	0.0157	0.0194	0.00280	0.103	0.313	0.00537
S-200	30M	0.0449	0.0342	0.00680	0.0152	0.0211	0.0182	**0.0150**	0.0151	0.0151	0.0171	0.00224	0.0892	0.290	0.00442
M-200	100M	0.0255	0.0144	0.00427	0.0123	0.0179	0.0151	0.0182	0.0117	0.0149	0.0142	0.00218	0.0329	0.191	0.00452
L-200	500M	0.0235	0.0117	0.00383	0.0114	0.0153	0.0133	0.0171	0.0108	0.0140	0.0158	0.00197	0.0307	0.182	0.00480
T-500	7M	0.0520	0.0367	0.00580	0.0112	0.0195	0.0153	0.0174	0.0138	0.0156	0.0148	0.00241	0.0910	0.280	0.00391
S-500	30M	0.0322	0.0237	0.00437	0.0129	0.0167	0.0148	0.0152	0.0126	0.0139	0.0129	0.00235	0.0867	0.268	0.00382
M-500	100M	0.0229	0.0126	0.00335	**0.00998**	0.0146	0.0123	0.0161	0.00947	0.0128	0.0103	0.00227	0.0294	0.172	0.00373
L-500	500M	**0.0213**	**0.0104**	**0.00323**	0.0108	**0.0131**	**0.0119**	0.0160	**0.00905**	**0.0125**	**0.00739**	**0.00170**	**0.0278**	**0.170**	**0.00322**

2D训练，3D也能使用

DPOT:Auto-Regressive Denoising Operator Transformer

Scaling测试

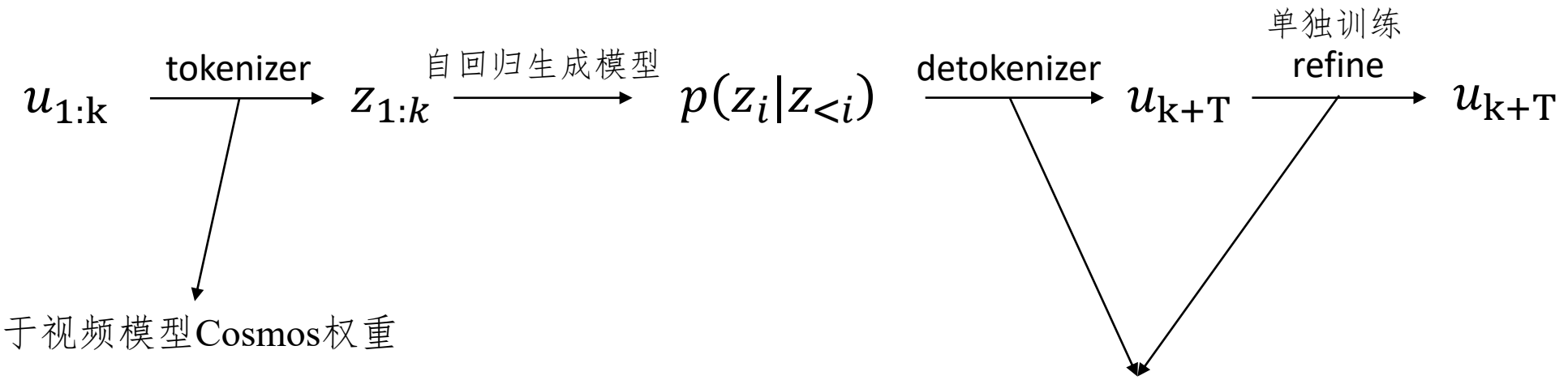




PhysiX

把physics变成token sequence，借鉴视频模型

$$u_{1:k} \rightarrow u_{k+T}$$



基于视频模型Cosmos权重

每种数据有不同的detokenizer和refine模块，并非Universal

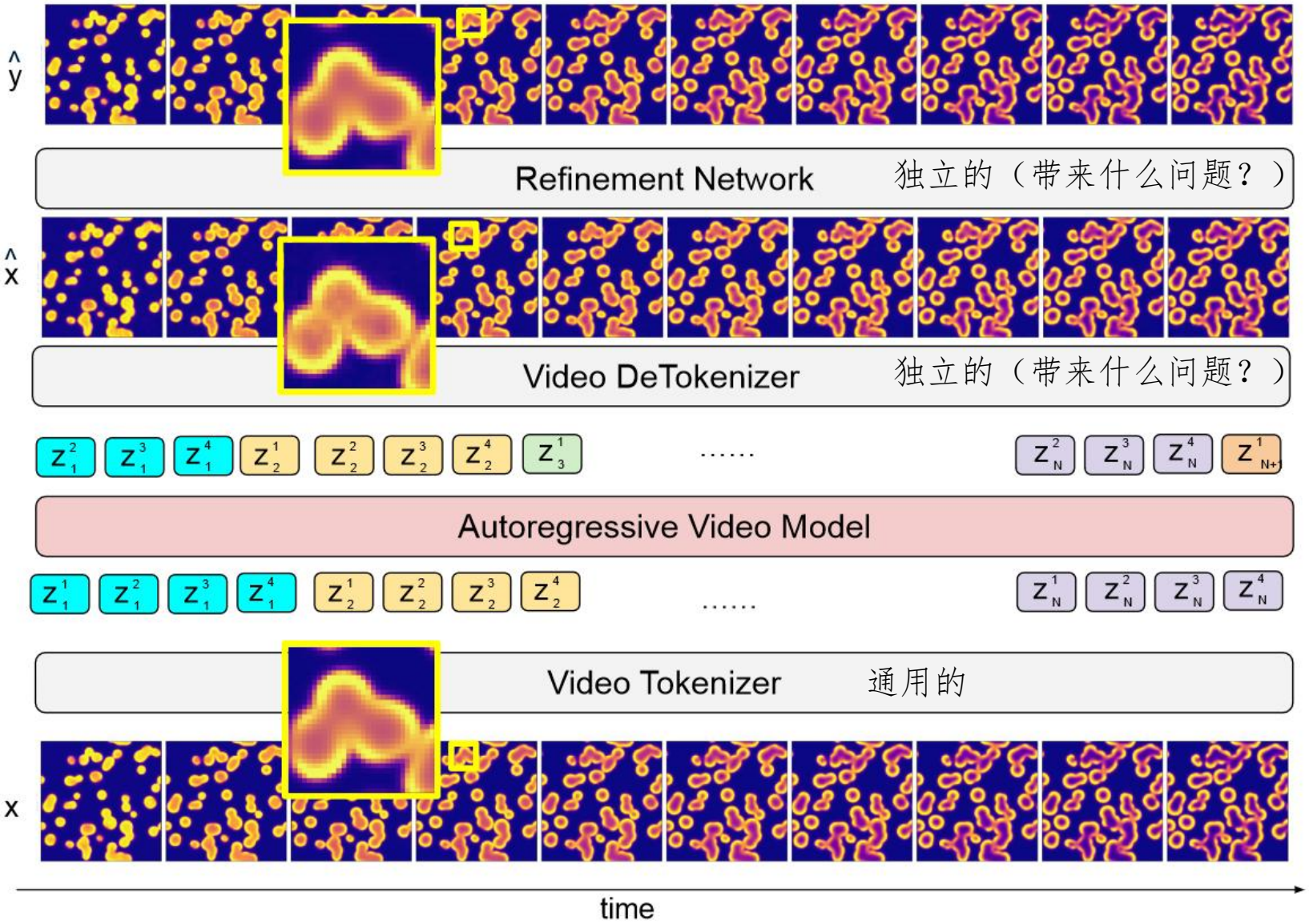
不同数据的缺失通道用2D张量padding

所有数据用通用的tokenizer encoder

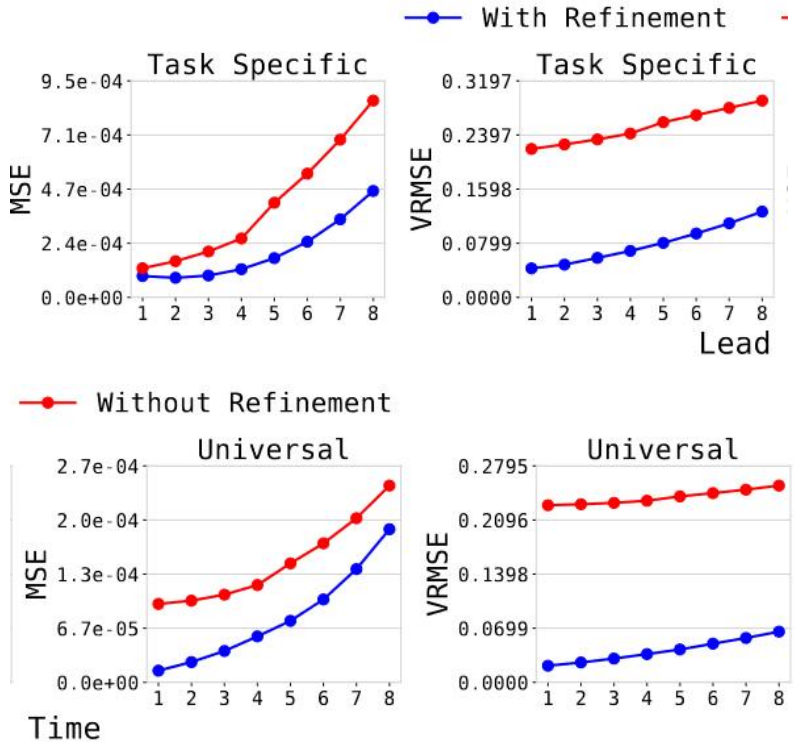
Refine模块暴露的问题：Tokenizer本身会带来误差，
视频可以接受，但PDE不一定，小尺度误差破坏结构

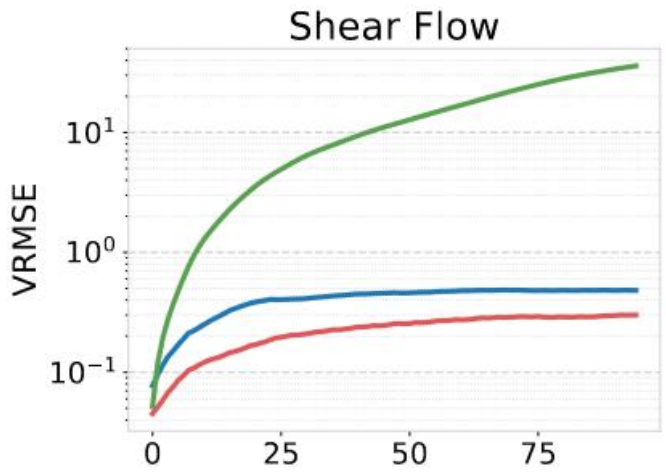
PhysiX

把physics变成token sequence，借鉴视频模型



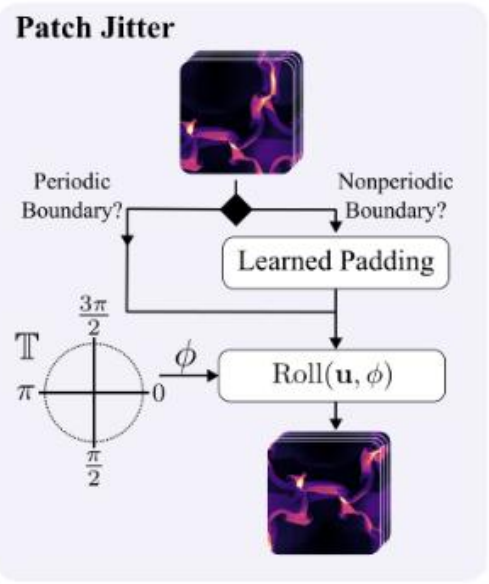
长时间预测和微调的效果较好

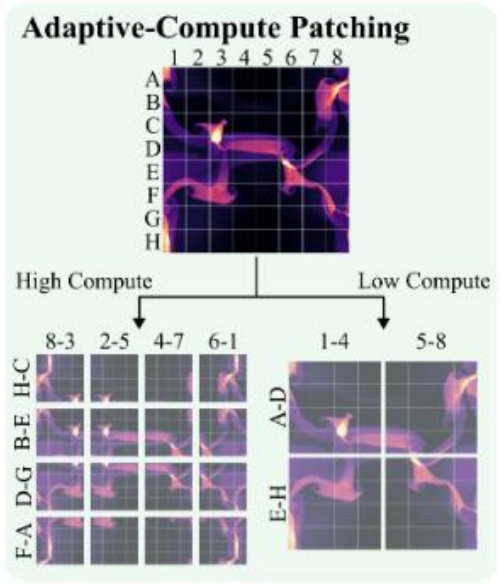


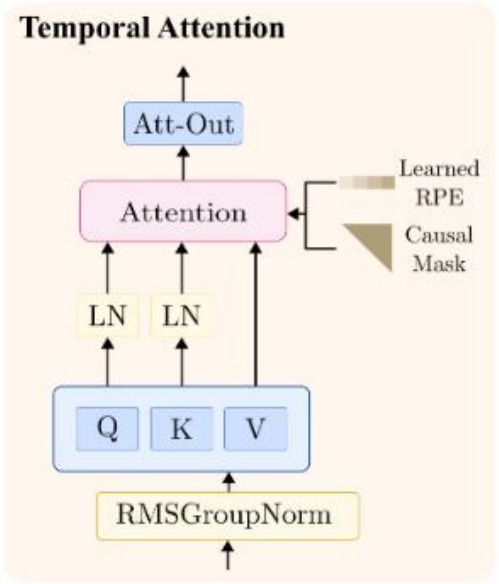


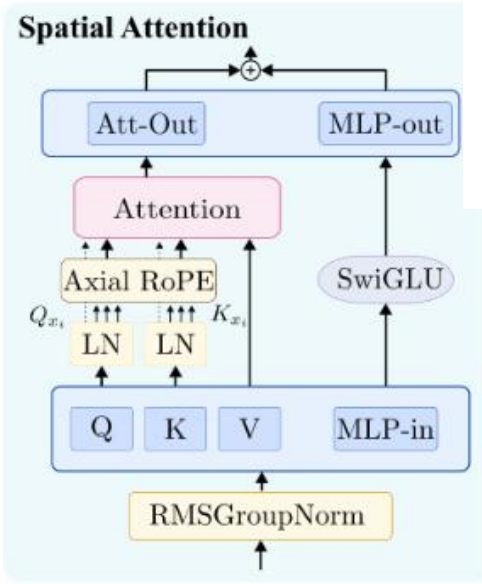
Walrus

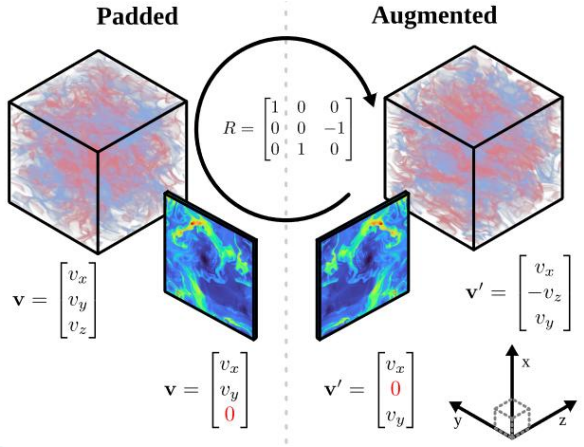
如何让token跨domain、dimension









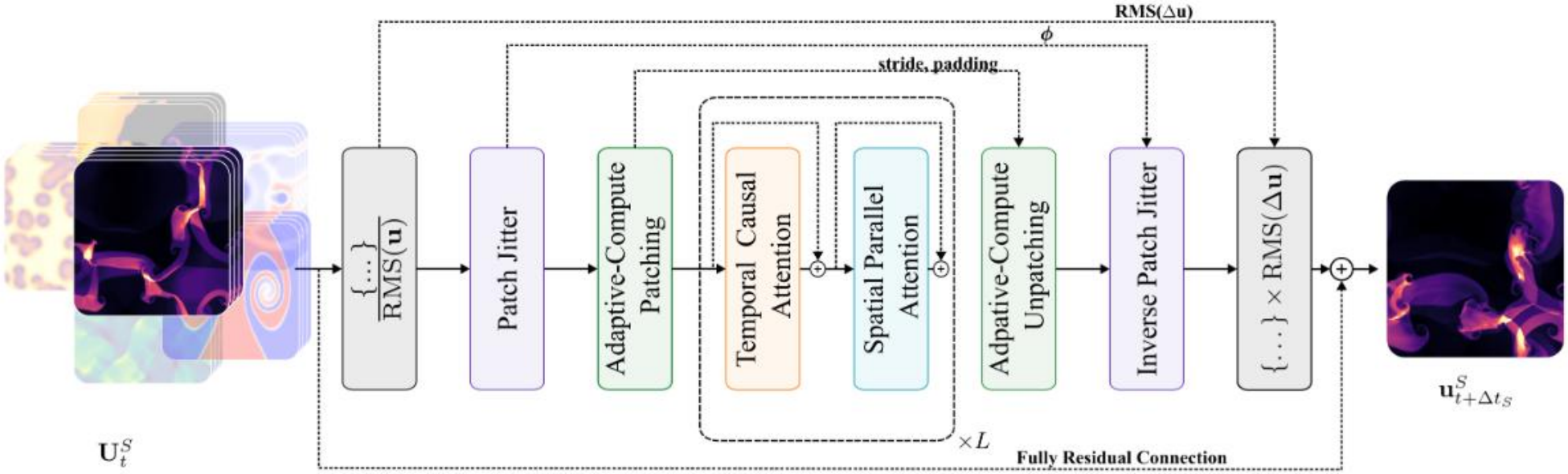


Patch Jittering:消除混叠

2D-3D 联合增强

自适应Token

时空分离Transformer



Walrus

Dataset	VRMSE: One-step				VRMSE: Avg $T \in [1:20]$				VRMSE: Avg $T \in [21:60]$			
	MPP-AViT-L	Poseidon-L	DPOT-H	**Walrus**	MPP-AViT-L	Poseidon-L	DPOT-H	**Walrus**	MPP-AViT-L	Poseidon-L	DPOT-H	**Walrus**
Biological and chemical behavior												
Active Matter	0.0157	0.0214	0.0476	**0.0057**	0.3195	0.3355	0.5272	**0.1262**	1.2481	1.3015	1.2867	**1.2451**
Gray-Scot	nan	0.0048	0.0061	**0.0001**	nan	0.0411	0.1354	**0.0278**	nan	0.1295	0.6567	**0.1268**
Acoustics and wave propagation												
Maze*	0.0337	0.0116	0.0126	**0.0099**	0.0797	0.0785	0.0350	**0.0345**	0.1390	0.2429	**0.0543**	0.0560
Inclusions	0.0432	0.0127	0.0199	**0.0089**	0.1362	0.0510	0.0500	**0.0430**	0.4940	0.1407	**0.1328**	0.1427
Discontinuous	0.0097	0.0035	0.0055	**0.0021**	0.0460	0.0255	0.0162	**0.0093**	0.2310	0.0890	0.0398	**0.0385**
Staircase	0.0026	0.0019	0.0017	**0.0005**	0.0053	0.0201	**0.0022**	0.0040	0.0096	0.0507	**0.0031**	0.0074
Astrophysical and geoscience applications												
Supernova (3D)	—	—	0.6417	**0.2462**	—	—	0.7366	**0.6673**	—	—	**0.9669**	1.0963
TGC (3D)	—	—	0.1002	**0.0466**	—	—	0.3482	**0.2889**	—	—	0.6809	**0.6197**
PlanetSWE	0.0035	0.0035	0.0013	**0.0004**	0.0307	0.0730	0.0081	**0.0046**	0.1213	0.3452	0.0317	**0.0207**
Plasmas												
MHD (3D)	—	—	0.4734	**0.0580**	—	—	1.0250	**0.6487**	—	—	1.3055	**1.2256**
Viscous fluids												
Rayleigh-Benard	0.0264	0.0215	0.0288	**0.0059**	0.4109	1.3819	3.4868	**0.0992**	0.7468	0.9586	1.2664	**0.6441**
Shear Flow	0.0071	0.0090	0.0162	**0.0012**	0.0377	0.0657	0.0772	**0.0146**	0.1814	0.2408	0.2880	**0.0810**
FBHarmonics	0.0104	0.0068	0.0031	**0.0005**	0.0420	0.0686	0.0525	**0.0189**	0.1785	0.2526	0.1546	**0.0603**
RT Instability (3D)	—	—	0.2165	**0.0565**	—	—	0.3191	**0.0496**	—	—	0.9321	**0.4776**
Inviscid fluids												
MultiQuadrantsP	0.0418	**0.0142**	0.0397	0.0194	0.2010	**0.0682**	0.0749	0.0720	0.6860	**0.2807**	0.3839	0.3408
MultiQuadrantsC	0.0432	0.0158	0.0468	**0.0112**	0.3050	0.0846	0.4801	**0.0587**	1.1011	0.6478	2.7573	**0.2823**
TRL (2D)	0.1707	0.1323	0.1601	**0.0831**	0.4796	0.4117	0.5134	**0.3393**	0.8879	**0.8441**	0.9316	0.8648
TRL (3D)	—	—	0.2238	**0.1588**	—	—	0.5753	**0.5216**	—	—	0.8106	**0.7839**
Non-newtonian fluids												
Viscoelastics	0.1030	0.0878	0.1398	**0.0295**	0.1578	0.1532	0.1989	**0.0373**	—	—	—	—

预训练分布内数据集表现最佳

零样本迁移效果佳

2D-3D增强提升下游未知任务性能

局限：未支持非均匀几何
确定性建模
2个decoder（2D/3D各一个）

Phaedra

什么样的tokenizer合适？

论文本身是重建任务，不是预测

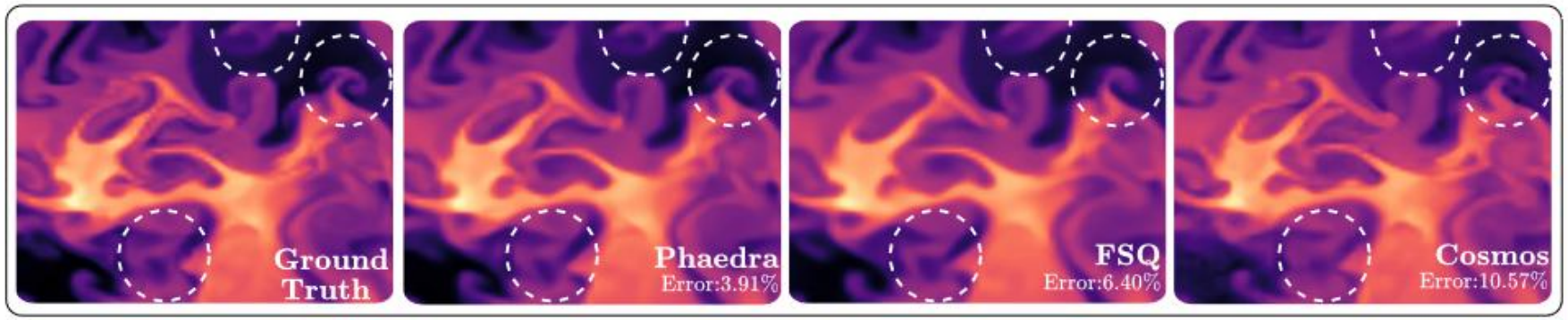
图像视频tokenizer合适？

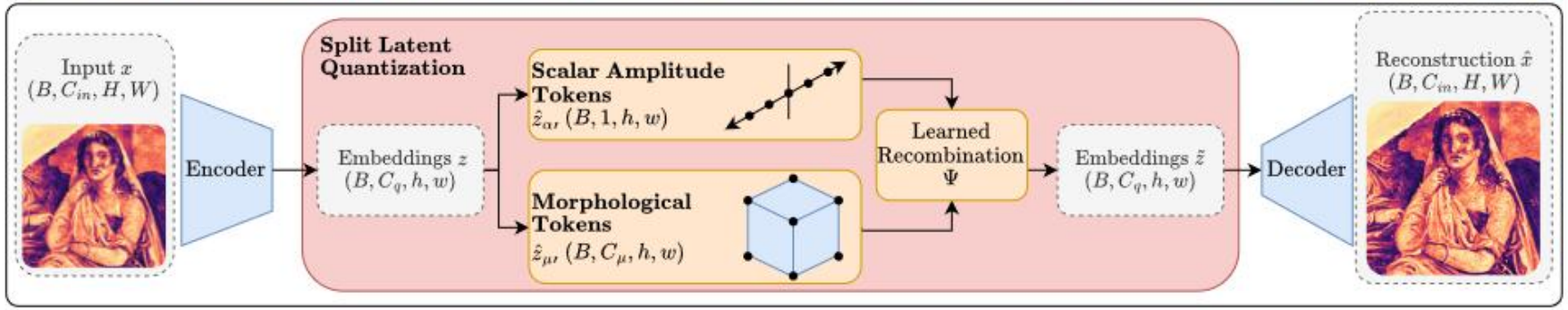
拆分

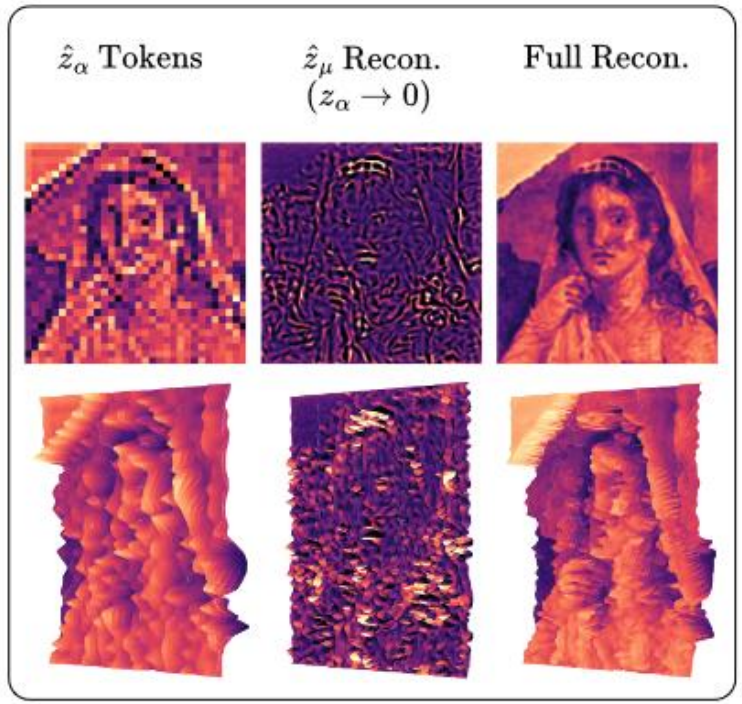
u → tokenizer → z → z_μ 形状隐变量，捕捉空间结构

z → z_α 幅值隐变量，捕捉能量密度

z_μ, z_α → $z = \Phi(z_\mu, z_\alpha)$ → decoder → u



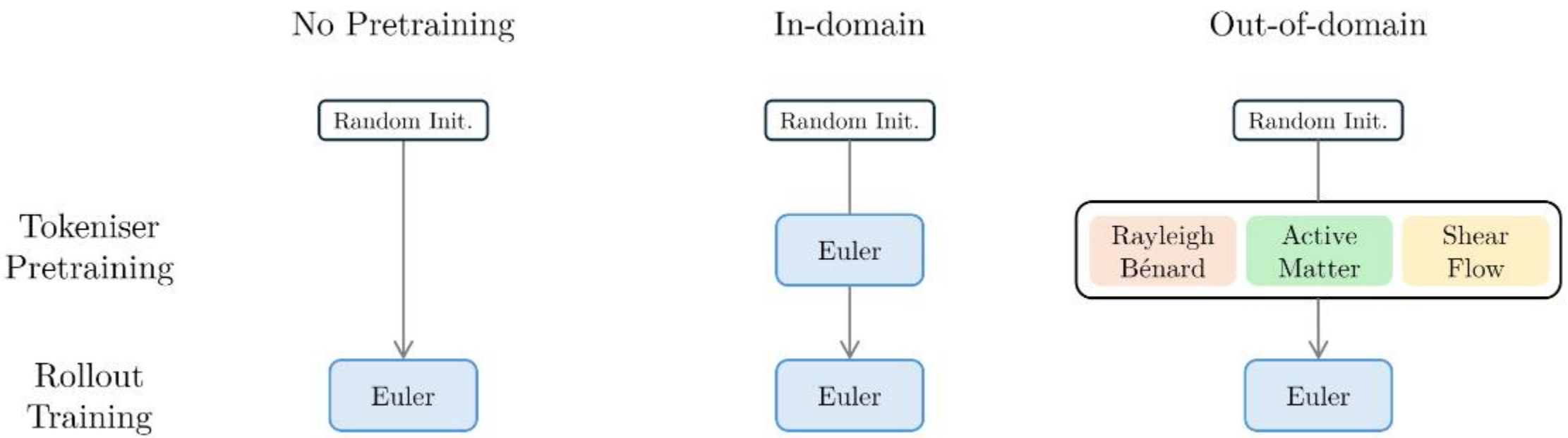




On the Value of Tokenizer Pretraining

Tokenizer的适用边界？

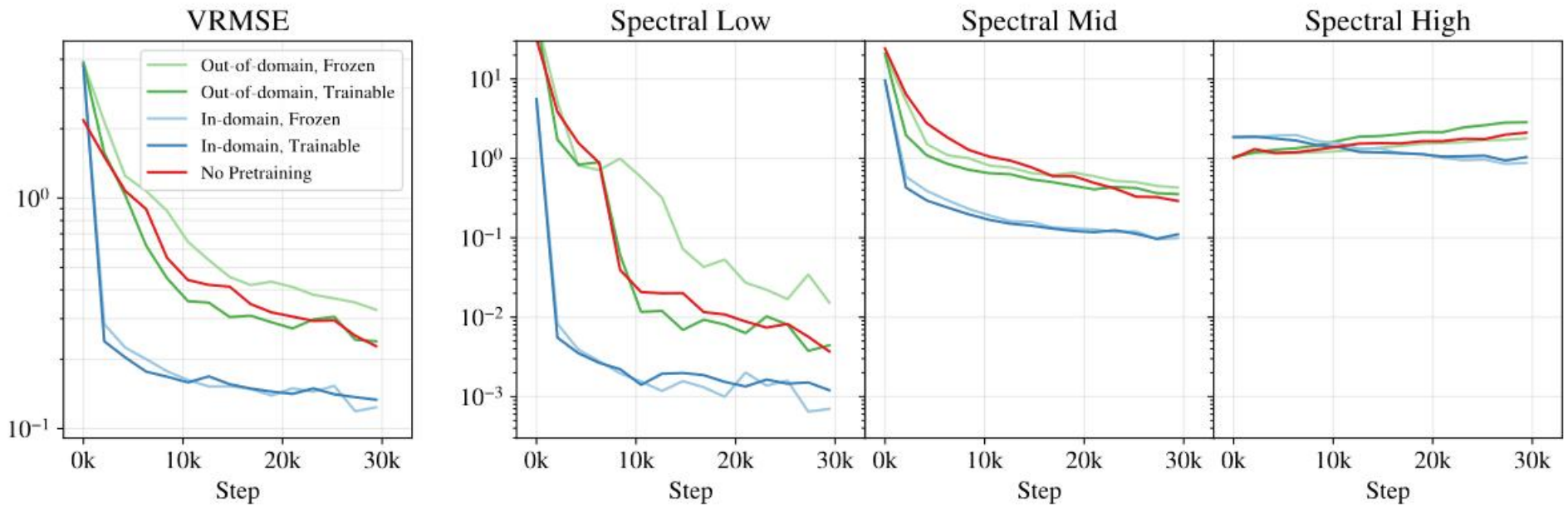
实验设置：三类对照



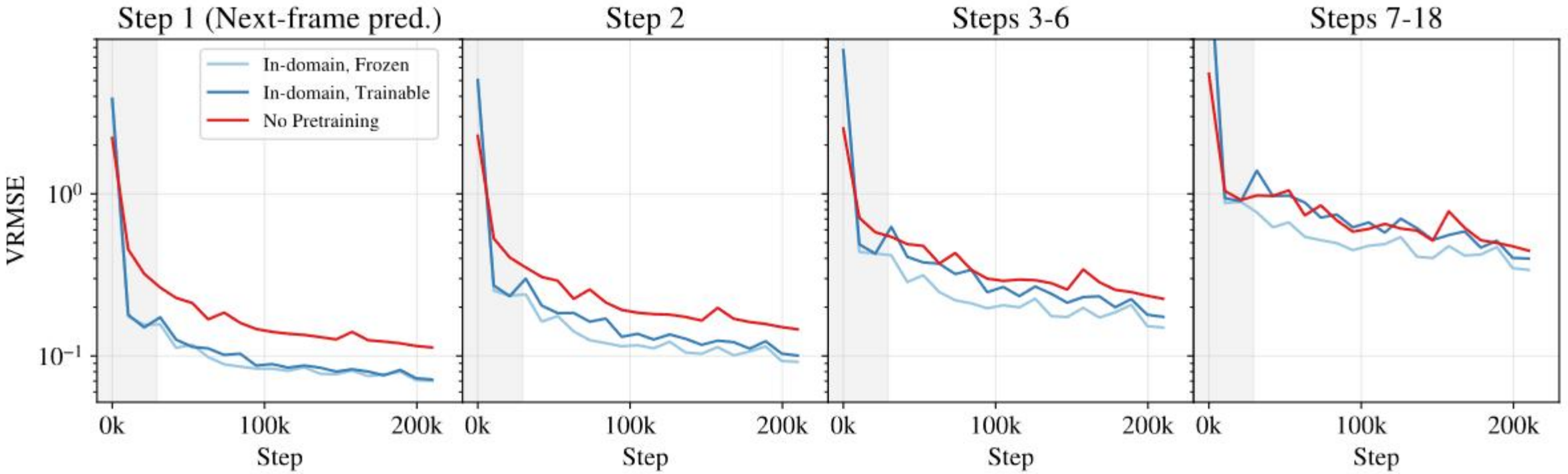
处理下游任务时，有两种处理方式：完全可训练的的tokenizer；大部分冻结的tokenizer

On the Value of Tokenizer Pretraining

	VRMSE	Spectral–Low	Spectral–Mid	Spectral–High
No pretraining	0.439	0.021	1.039	1.408
In-domain	0.158	0.001	0.167	1.381
In-domain	0.162	0.002	0.190	1.545
Out-of-domain	0.355	0.012	0.642	1.659
Out-of-domain	0.643	0.544	0.801	1.229



On the Value of Tokenizer Pretraining



领域对齐是预训练收益的决定性因素

误差和空间尺度强相关；高频小尺度结构仍是难点。

域内预训练之后冻结 Tokenizer 大部分权重，简单有效的正则手段，显著提升长时序仿真滚动稳定性。

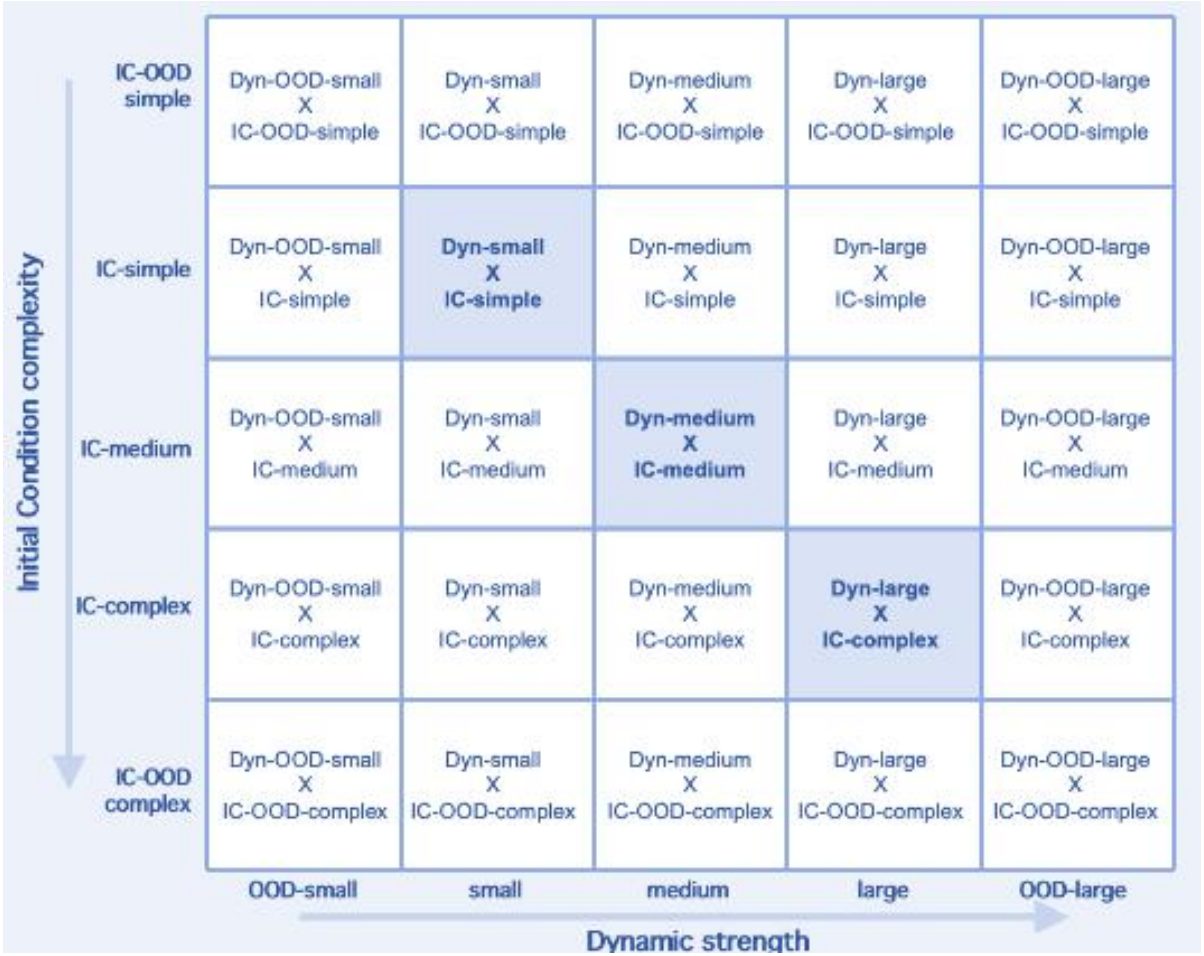
Takeaway

	DPOT	PhysiX	Walrus	Phaedra	Tokeniser Pretrain
原始输入	continuous fields	fields	fields	single physical field	fields
token 类型	patches	discrete	adaptive patches	discrete 拆分	adaptive patches
dynamics backbone	Fourier mixer	4.5B Transformer	1.3B space-time Transformer	—	100M space-time Transformer
预测对象	next field	next token	Δu	reconstruction	next field
稳定性机制	denoising	refinement	patch jitter	high-fidelity encoding	tokenizer pretraining
核心问题	怎么 pretrain	怎么表征	怎么跨域且稳定	什么样的token	representation 要不要先学

Takeaway

核心问题	代表工作	思路
Multi-PDE pretraining	MPP, DPOT, Poseidon	shared dynamics representation
PDE-aware conditioning	PDEformer, Unisolver, PROSE	equation / BC / coefficient 作为输入
Physics data scaling(数据)	The Well	heterogeneous simulation corpus
Tokenization	PhysiX, Phaedra	compress physical fields into tokens
Representation pretraining	Tokeniser Pretraining	decouple representation and dynamics learning
Cross-domain modeling	Walrus	2D/3D + heterogeneous physical systems

Do Physics Foundation Models Learn Generalizable Physics?



系统性物理工况偏差：在扩散类方程表现更好、波动更差，预训练求解PDE偏好

时域偏差：短时预测准不等于长序列稳定

泛化能力并不普适，条件化：动态尺度OOD难于IC-OOD

扩充数据只能局部改善，不能解决泛化，损害简单情况性能

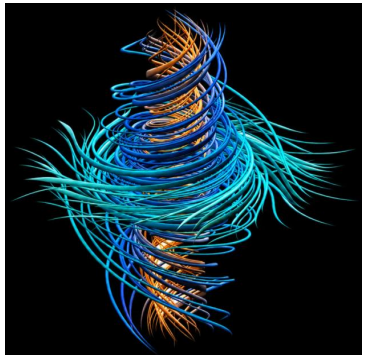
预训练只能带来偏好，不是通用物理先验，scaling很条件化

不同的Foundation Model缺陷来源不同，需要针对性改进，没有通用修复方案

物理基座模型：实现痛点

Data

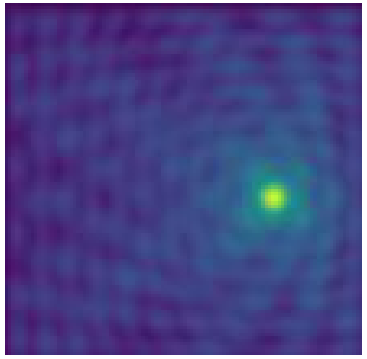
数据昂贵、异构



大规模、高质量、
统一协议的数据集

Representation

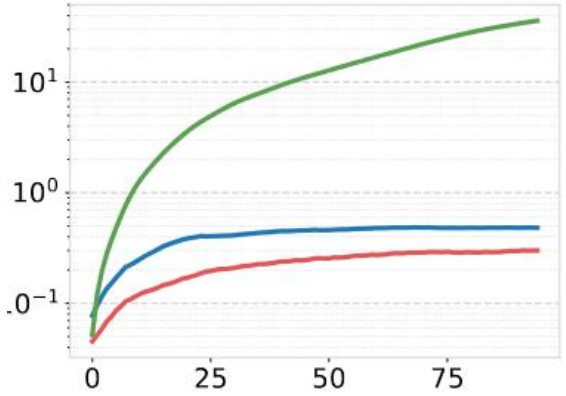
2D、3D、tensor、vector



通用、有效的表征
多尺度数据的统一

Dynamics

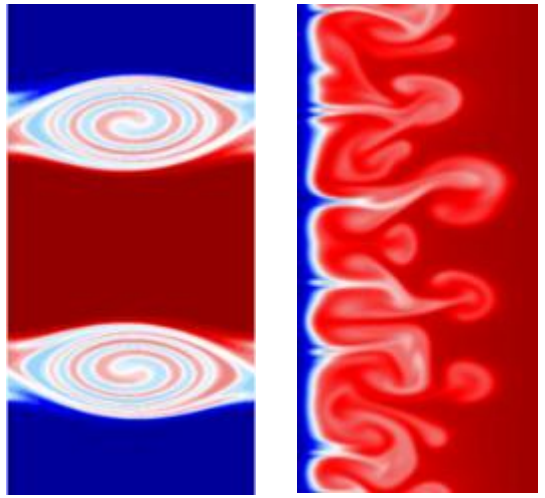
长时累积误差、多尺度



长时预测

Generalization

泛化到未见物理？



通用的物理知识
跨物理问题

理想中的 Physics Foundation Models

多模态输入：不只可以是物理场的前几帧，也可以是PDE，IC, BC，甚至是文字、图片

多种能力：不只可以预测，也可以重构、逆向设计、控制

精准的长时预测、跨物理能力：长时预测物理场演化，理解通用物理先验，能支持快速仿真、甚至控制

每一个环节都很困难